

PSU ECE Capstone Team 4 Contract

Contact Information

Team Members

Team Member Name	Phone	Email
David Mazhnikov	503-676-7447	dmazhn2@pdx.edu
Ethan Budde*	503-990-3339	budde@pdx.edu
Roman Sulza	971-352-2659	rsulza@pdx.edu
Clarker Ren	503-421-7889	clarkere@pdx.edu
Hussein Ashkanani	503-488-0470	hashk2@pdx.edu

If you have a project coordinator or leader, please mark them with an asterix ().*

Industry sponsor

Organization and Name	Phone	Email
Healthy Buildings Research Laboratory - Dr. Elliott Gall	(503) 725-2878	elliott.gall@pdx.edu

Faculty Advisor

Name	Phone	Email
Joshua Méndez	(503) 725-3000	joshua.mendez@pdx.edu

Project Objective

This is the “north star” for the project. What is your team’s objective, as voiced by the team? It should obviously align with the capstone project as described by the industry sponsor, but say it in your own words. Also add objectives your team would like to have, e.g., grades, skills learned, experiences had, etc.

Our team identified three primary goals to keep close to every aspect of design and decision making:

- 1. A commitment to accuracy - as we are designing highly performant experimental equipment, a commitment to high quality work and consistent testing of tools is required, in order to keep the system highly accurate in detection and communication.**
- 2. A consistency in organization and clarity - Clean, tabulated work is the most effective means of sharing a workload among team members, and is core to our goals of continuous team improvement.**
- 3. A cogent system of documentation - To support our goals of accuracy, organization, and clarity, we will leverage a set of documentation tools to notate our process methodology.**

Member Skills, Strengths, and Areas of Practice

This table should be derived from your team canvas and team discussions, and then used throughout the project in determining task ownership, team processes and helping each other develop skills. We don’t like “weakness” as much as “areas where one needs practice”.

Teammate Name	Key areas of strength to leverage in the project	Key areas of practice and development in the project
Ethan Budde	Coding, Hardware, Outlining & Documentation, Project Lead	Time and documentation management
Hussein Ashkanani	Coding and Outlining / Documentation	Hardware & Physical prototyping
Clarker Ren	Coding & Firmware	Hardware & MCAD Design
David Mazhnikov	Hardware Design, Physical Prototyping, and Circuit Test/Debug	Coding & Integration
Roman Sulza	Hardware and Physical / Firmware	Testing and debugging

Meetings

Team Meetings

- **Frequency:** At least twice a week
- Proposed days and times:
 - Gaggiuino Team: Wednesday 1:00 PM
 - Gaggi Mate Team: Tuesday/Thursday 10:00 AM, optional Friday 11:00 AM ? based on when2meet
 - Full ECE/ME Team Meetings: Fridays 1:30 PM
- **Platform:** Primarily in person w/ accompanying Discord/Zoom.
- **Expectations:** Efficient conversation of relevant topics, equal and fair respect for ideas and their representation, and a respect for the time of all attending parties.
- **Agenda:** Will be defined at the start of each meeting and include status updates, technical blockers, upcoming deadlines, and task assignments.
- **Prep:** Work done for meetings will largely be doled out week by week by relevant parties working on those tasks.
- **Attendance:** Nonmandatory, but highly encouraged. Repeated absence without communication may result in follow-up discussions.
 - Steps may be taken if members routinely miss meetings.
- **Participation:** Members are expected to speak when the meeting discusses problems or assignments relevant to their work within the project.
- **Note-taking (minutes):** 1-2 minute takers.
- **Other expectations:** Clear actions should be identified before ending each meeting.

Team meetings with Industry Sponsor & Faculty Advisor

- Frequency: Up to twice a week
- Proposed days and times: In-line with CE meetings
- Platform: Zoom
- Expectations: These meetings are intended to provide progress updates, propose ideas, and receive feedback from the sponsor. The team is expected to come prepared with clear questions or updates. This asynchronous system gives advisors and sponsors the most efficient connection to team meetings.

Communications

Preferred Method of Communication

List your communication tools and if there is anything important about how you use them

- **Discord**
 - **Cross channel communication with EE team.**
 - **Daily communication and task coordination.**
- **Zoom**
 - **Meetings with sponsors, advisors, remote participants.**

Information Architecture

Define how information will be shared, your project management tool, and any other technology tools anticipated (e.g., GitHub, google drive).

- **GitHub**
 - **Version control for firmware, code, and documentation.**
- **Google Drive**
 - **Shared documents, reports, schedules, and presentation materials**
- **Discord**

Team Working Agreements and Expectations

Guiding principles of how we work together

Are there any thoughts or values that guide how your group works together?

The team values clear communication, mutual respect, and accountability. Members are encouraged to ask questions early.

How we make decisions

How does your group make decisions?

Group decision making will be done through a committee of relevant members for each decision taking a vote on actions to take. Given a choice between x options, each option is given it's time to be analyzed for upsides and downsides, and compared against each other quantitatively. Given an outline of what the decision "means" (what the tradeoffs of the vote are), each relevant party votes, and the majority is respected. In the event of a tie, a respected external party (likely an EE student or non-relevant ECE student) is given context, and asked to act as final arbiter.

Approach to conflict

Consider productive conflict, interpersonal conflict and when to elevate an issue to the Faculty Advisor or to the Capstone Coordinator.

Conflict will be mitigated through group decision making, giving each individual (with relevant attachment to the problem at hand) an equal platform to voice their ideas. In the event of a member's further dissatisfaction with the result, two more approaches are taken to balance:

- 1. Rationalization and reassurance to conflicted members that the nature of the decision has no personal basis, rather ONLY to choose the best option for the development of the project.**
- 2. Development of a potential compromise within the solution, such that all relevant members are decently satisfied with the outcome.**

Further conflict resolution will be done through the assistance of the Faculty Advisor and Capstone Manager.

Project Standards

What is a realistic level of quality that the team wants to hold themselves to? Any strategies to fulfil these standards? Also, verbally discuss your grade expectations, but please do not write them down here.

Our primary goal when designing and developing any part of the project is to work towards a high degree of accuracy first and foremost. We reached this goal after discussing how, as we are developing scientific equipment, a high degree of repeatable, detectable, and quantifiable accuracy of our machines was highly important for the rest of development to continue unabated.

How we will hold each other accountable

Regular mechanism for tracking, consider balanced participation, adherence to deadlines, approach to feedback. How you will candidly but kindly deal with team members who aren't fulfilling team expectations.

Accountability will be maintained through regular check-ins during meetings. If a member is struggling to meet expectations, the issue will be addressed respectfully through direct communication and, if needed, adjustment of workload.

How we will give and receive feedback

A high-performance team uses continuous improvement. This takes discussion and feedback on a regular basis. Giving and receiving feedback, aka “coaching”, is an important skill to master and use to enhance your capstone experience and performance. This can be difficult. Here are some best practices as a reminder:

Feedback will be given respectfully, focusing on behaviors, deliverables, and technical outcomes rather than personal traits. Team members are encouraged to ask clarifying questions, share their perspectives, and work collaboratively toward solutions.

Team Contract Signatures

By typing in your name, you agree:

- a) I participated in formulating the standards, roles, and procedures as stated in this contract.*
- b) I understand that I am obligated to abide by these terms and conditions.*
- c) I understand that if I do not abide by these terms and conditions, I will suffer the consequences as stated in this contract.*

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